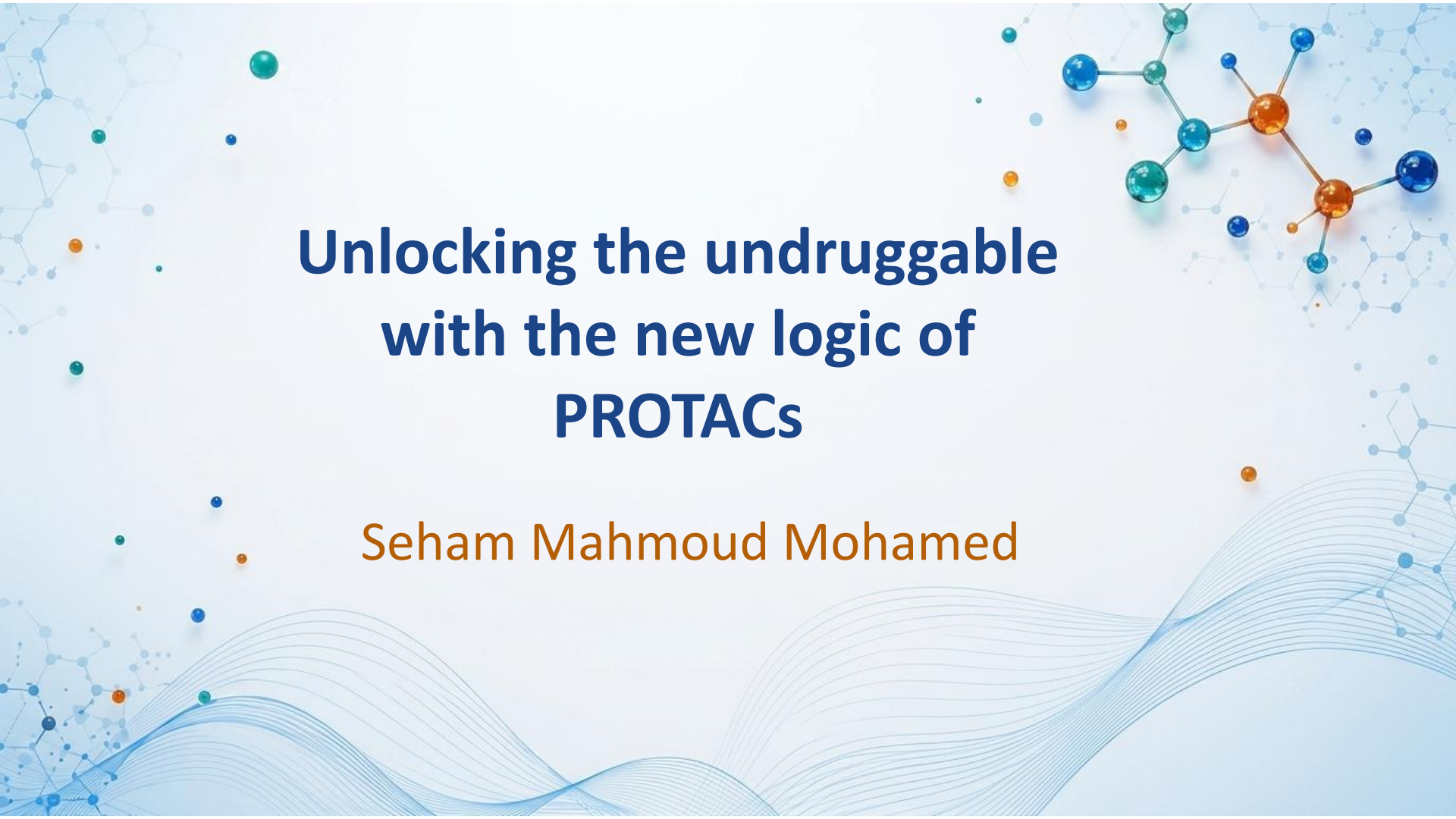


The background features a light blue gradient with abstract molecular structures. In the top right, there is a cluster of spheres in blue, green, and orange connected by lines, resembling a chemical molecule. The bottom of the image is decorated with several thin, wavy blue lines that sweep across the width of the frame. Scattered throughout are small, semi-transparent spheres in blue, green, and orange.

Unlocking the undruggable with the new logic of PROTACs

Seham Mahmoud Mohamed

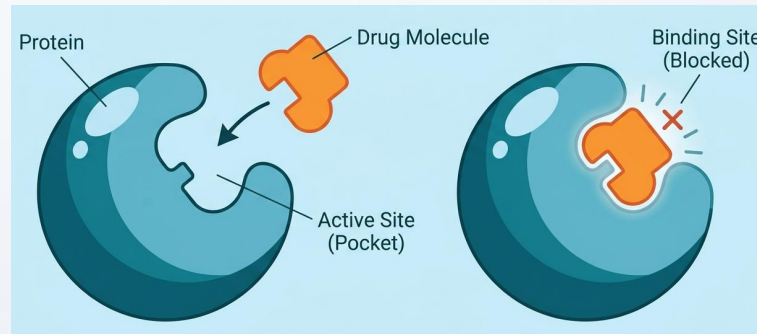
What does "druggable" mean?

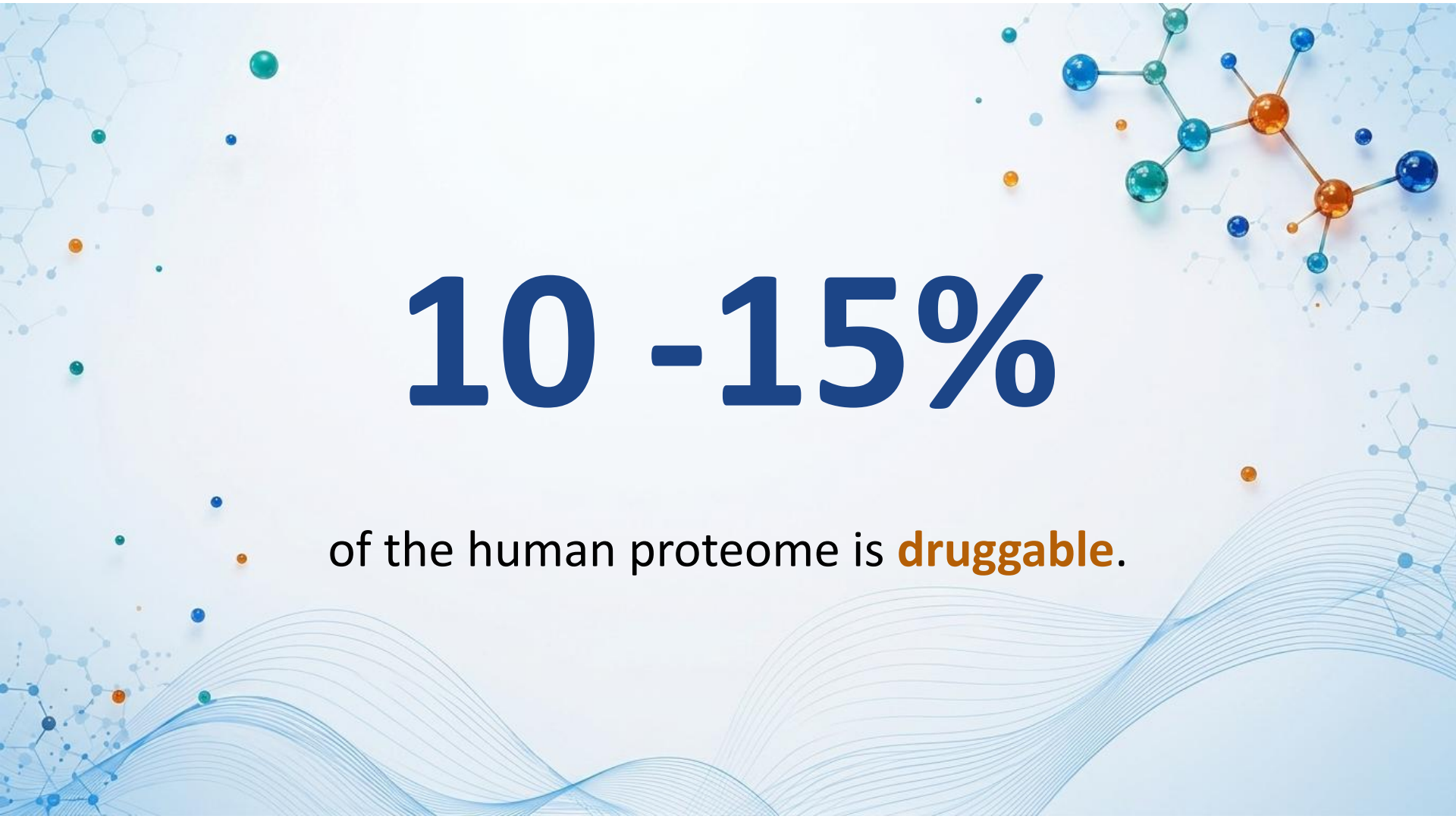
- A protein is considered **druggable** if a small molecule can bind to it with sufficient affinity and specificity to alter its function.

Examples:

Enzymes

• **Many receptors**



The background features a light blue gradient with decorative elements. In the top corners, there are molecular structures represented by spheres (blue, green, orange) connected by lines. At the bottom, there are wavy, horizontal lines in a light blue color.

10 - 15%

of the human proteome is **druggable**.

The undruggable proteins

They are proteins that are difficult to target by traditional drugs because they:

- Lack well-defined ligand-binding **pockets**.
- Possess **large, flat** protein–protein interaction (PPI) **interfaces**.
- Are intrinsically disordered proteins (**IDPs**).
- Are **non-enzymatic** proteins without catalytic active sites.
- Adopt dynamic or **transient conformations**.

Examples:

Transcription factors - scaffolding proteins - regulatory proteins.

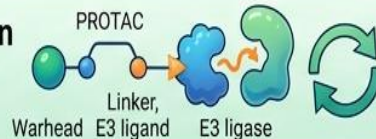
Evolution of Therapeutic Approaches



Limited Target Space



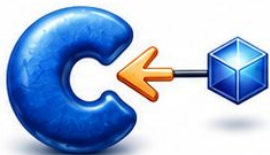
**Targeted Protein
Degradation
(PROTACs)**



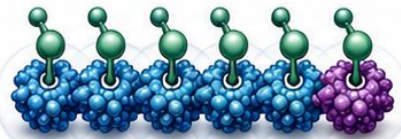
The background features a light blue gradient with abstract molecular structures. In the top right, there is a cluster of spheres (blue, green, and orange) connected by lines, resembling a chemical structure. The bottom of the image is decorated with several thin, wavy, light blue lines that sweep across the width of the frame. Scattered throughout the background are small, semi-transparent spheres in blue, green, and orange, some of which are part of larger, fainter molecular-like structures.

Targeted protein degradation

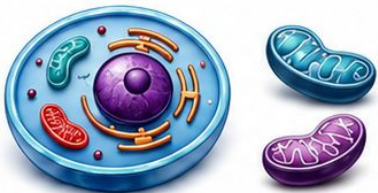
TRADITIONAL INHIBITOR



Block activity



Require continuous occupancy

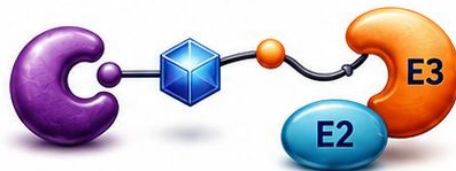


Protein remains in the cell

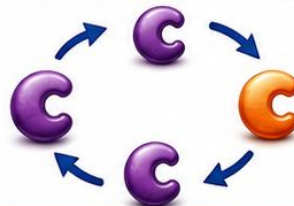


More susceptible to resistance

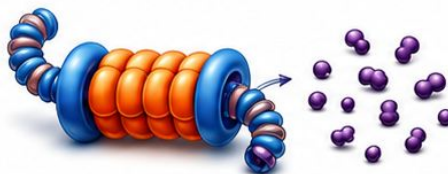
TARGETED PROTEIN DEGRADATION (TPD)



Removes protein



Catalytic mechanism

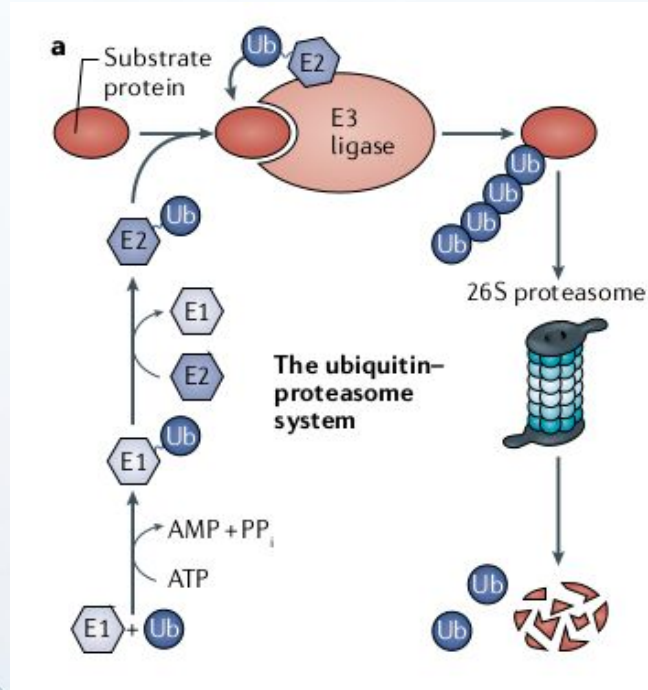


Protein degraded

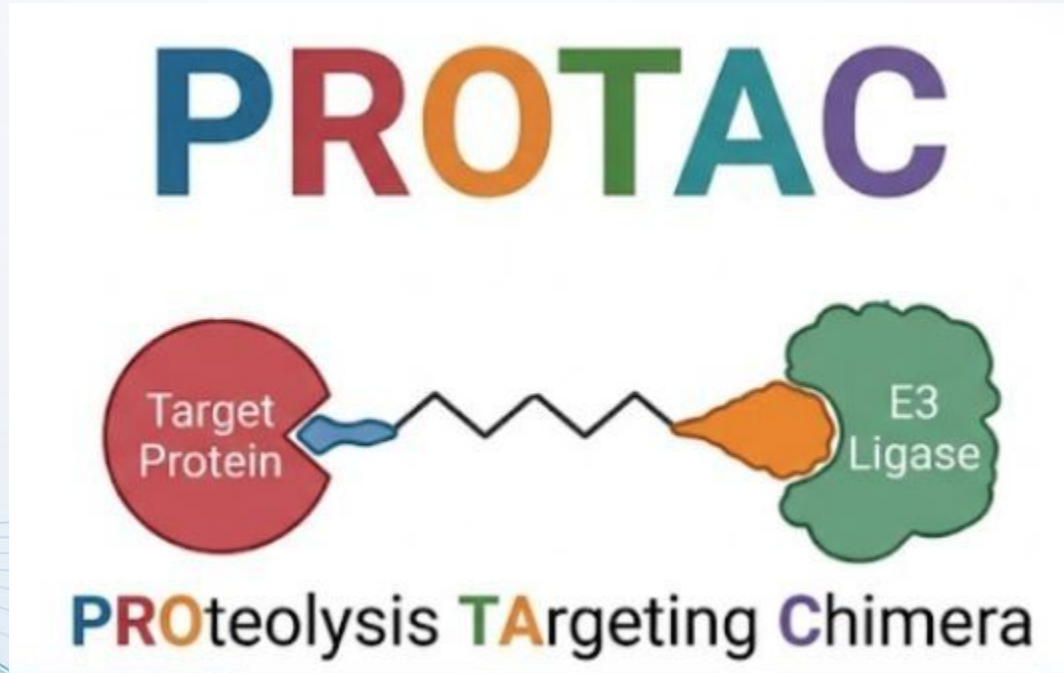


May overcome some resistance mechanisms

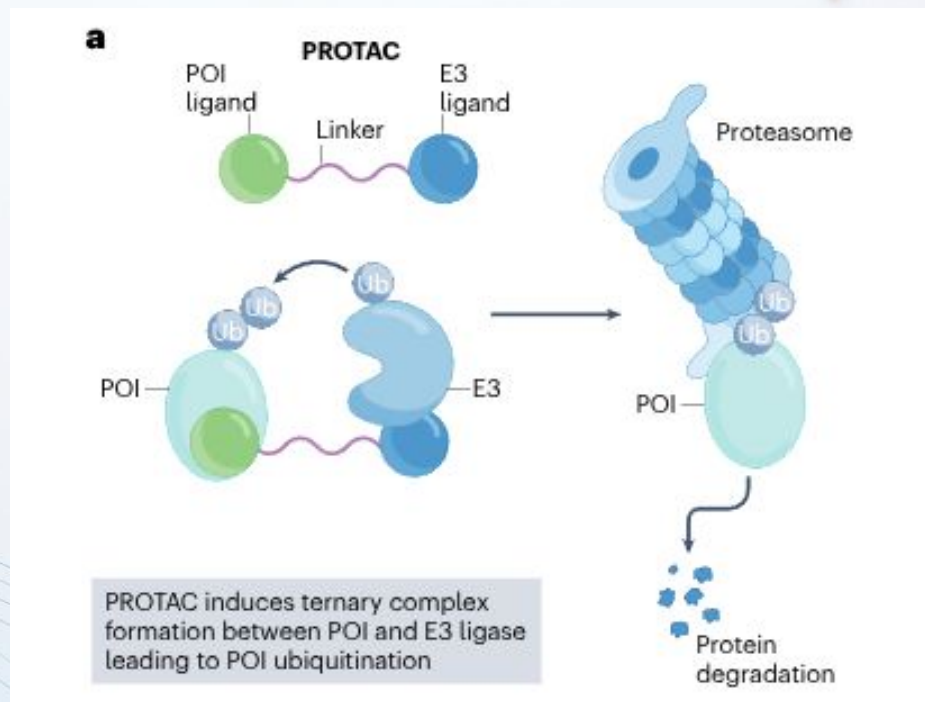
The Ubiquitin-Proteasome system



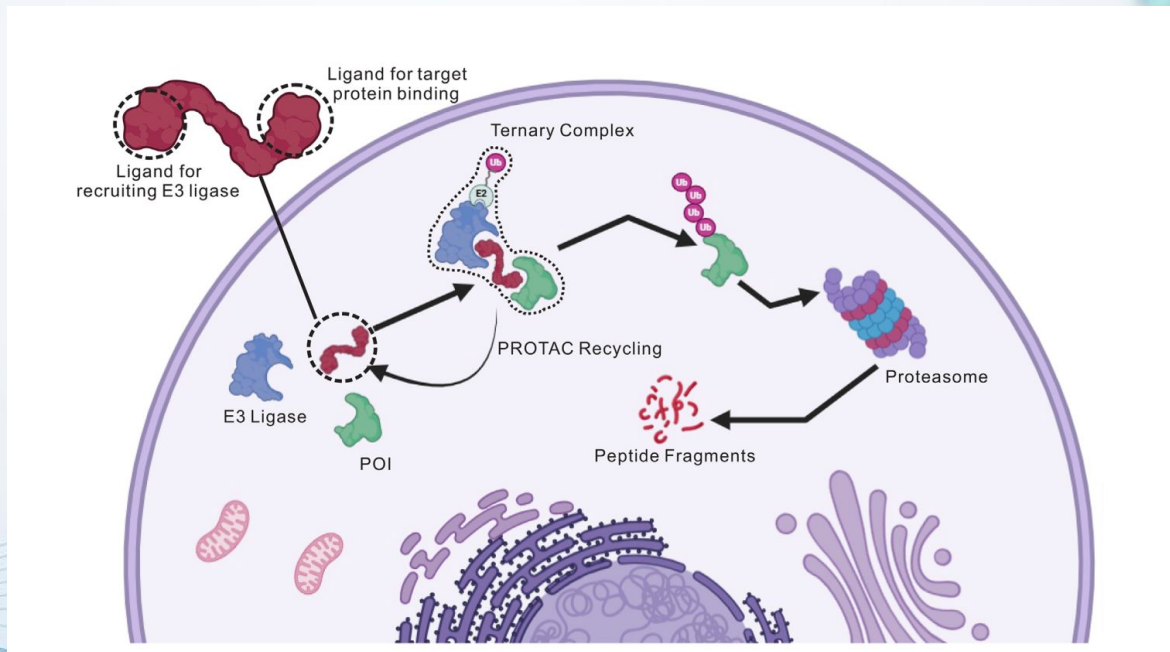
What is meant by PROTAC?



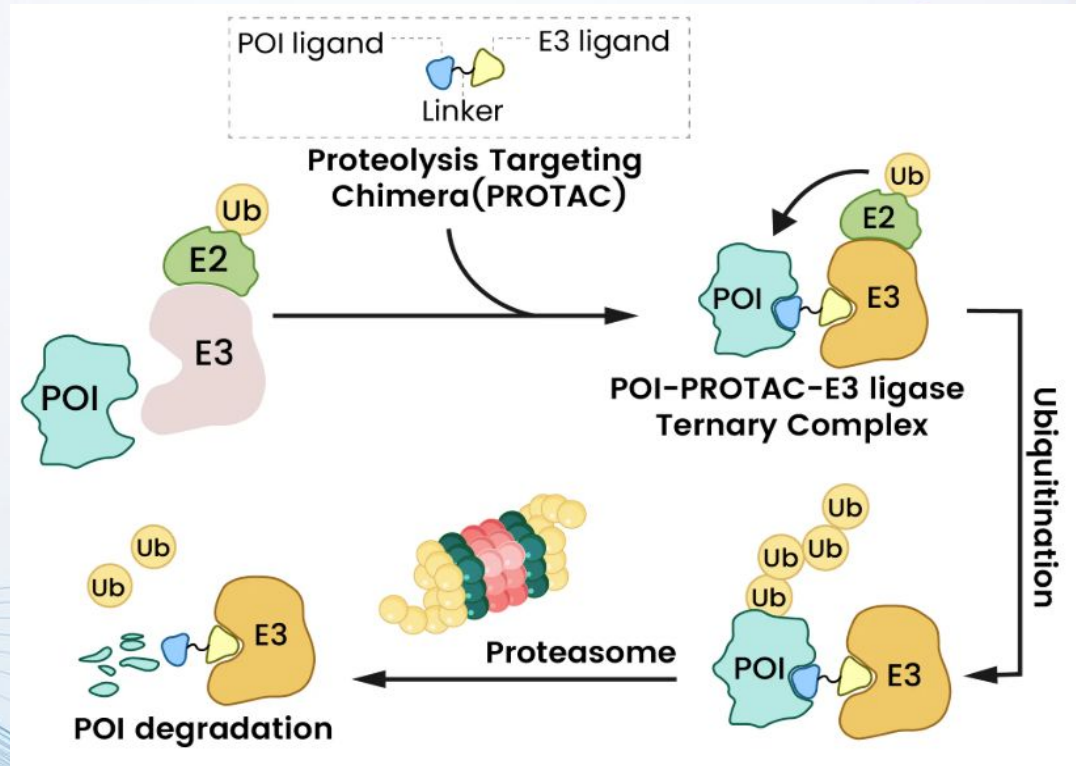
Components of PROTAC



Mechanism of action



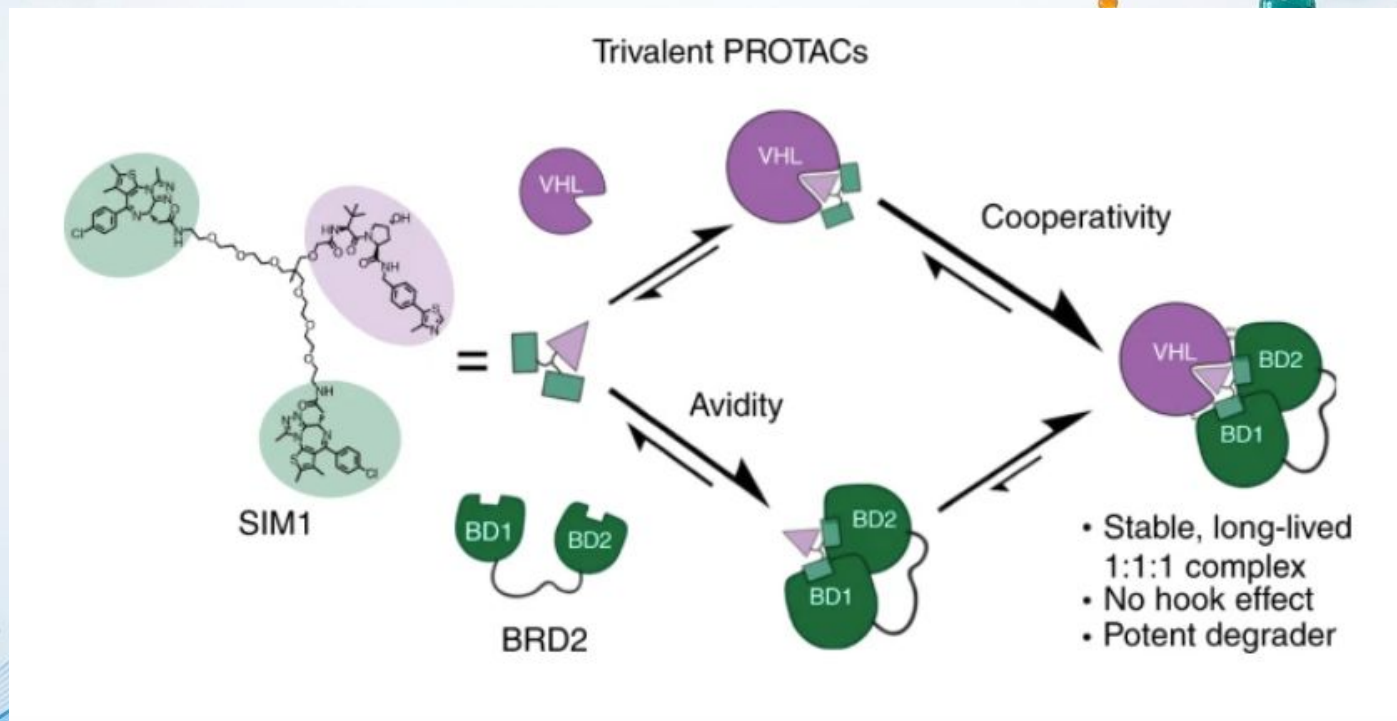
Fan, et al., *MedComm*, 2025, 6(10).



The unique mechanism of PROTAC

- ✓ **Event-driven** pharmacology.
- ✓ **Catalytic** mode of action.
- ✓ One PROTAC molecule can degrade **multiple target** proteins.
- ✓ Protein is **eliminated** rather than temporarily inhibited.

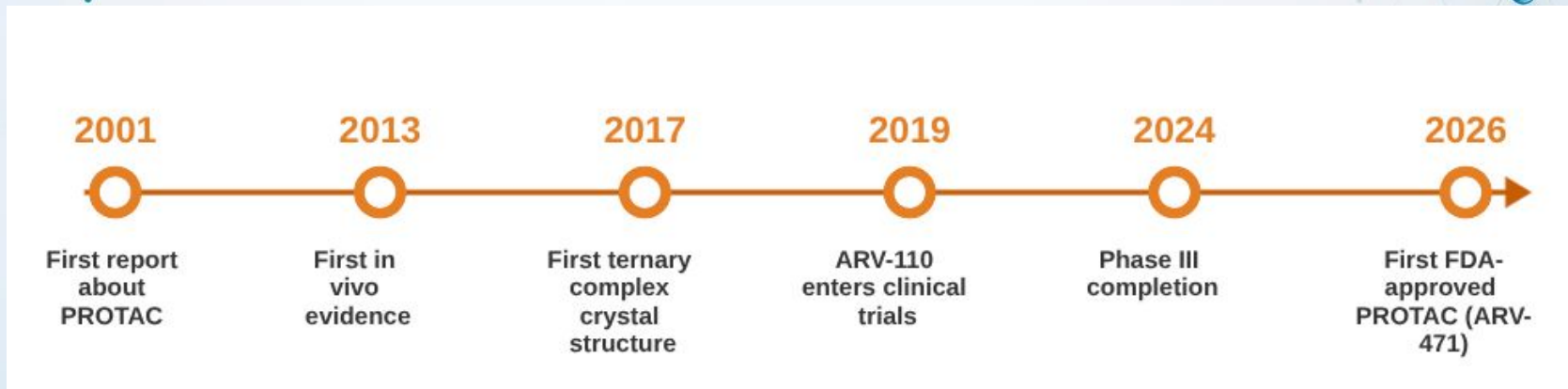
Trivalent PROTACs



PROTACs vs other conventional therapeutics

Feature/ capability	Small molecule inhibitors	Monoclonal antibodies (mAbs)	Gene-based medicines	PROTAC protein degraders
Mechanism of action				
Occupancy-driven (binds & inhibits)	Yes	Yes	No	No
Event-driven (catalytic degradation)	No	No	Possible	Yes
Sub-stoichiometric activity (catalytic)	No	No	Possible	Yes
Target engagement				
Eliminates disease-causing proteins	Indirectly	Indirectly	Yes	Yes
Disrupts Scaffolding/nonenzymatic function	Weak/no	Possible	Yes	Yes
Targets “undruggable” proteins	Rarely	Rarely	Yes	Yes
Targets intracellular proteins	Yes	No	Yes	Yes
Targets cell surface proteins	Possible	Yes	Limited	Possible
Pharmacokinetics/administration				
Broad tissue penetration	High	Low	Low/variable	Moderate
Systemic delivery	Yes	Yes	Limited/challenging	Yes
Oral bioavailability	High (many)	No	No	Possible/developing
Route of administration	Oral/Injection	Injection	Injection/complex	Oral/injection
Development & practicality				
Ease of manufacturing	Relatively easy	Complex	Very complex	Moderate/developing
Ease of development (drug discovery)	Challenging	Established	Complex	Emerging/potential
Potential for high dose dependence	Yes (often)	Yes	Variable	Low (catalytic)

Timeline of PROTAC



PROTAC-1

Protacs: Chimeric molecules that target proteins to the Skp1-Cullin-F box complex for ubiquitination and degradation

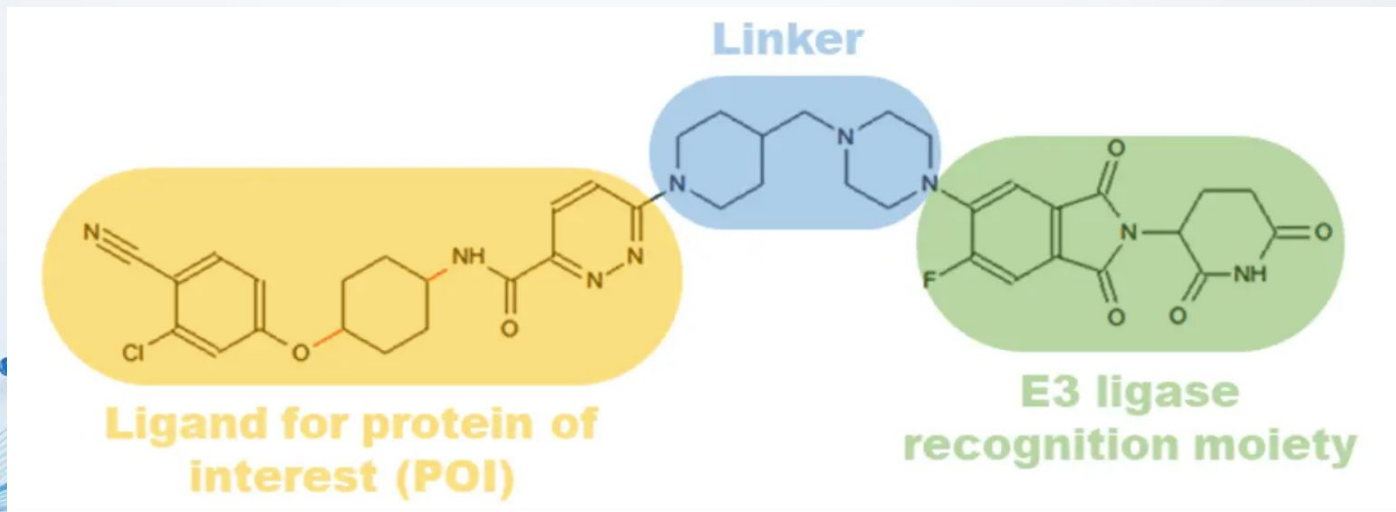
[Kathleen M Sakamoto](#) ^{*,†,‡}, [Kyung B Kim](#) [§], [Akiko Kumagai](#) [†], [Frank Mercurio](#) [¶], [Craig M Crews](#) [§], [Raymond J Deshaies](#) ^{†,||,‡}

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PMCID: PMC37474  IF: 9.5 Q1 PMID:

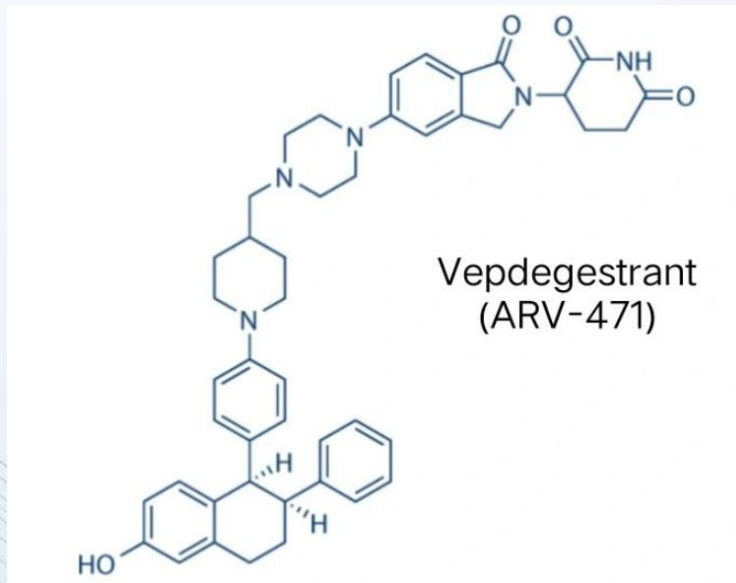
ARV 110

a PROTAC **Androgen** Receptor Degrador, in metastatic castration-resistant **prostate** cancer



The first FDA-approved PROTAC

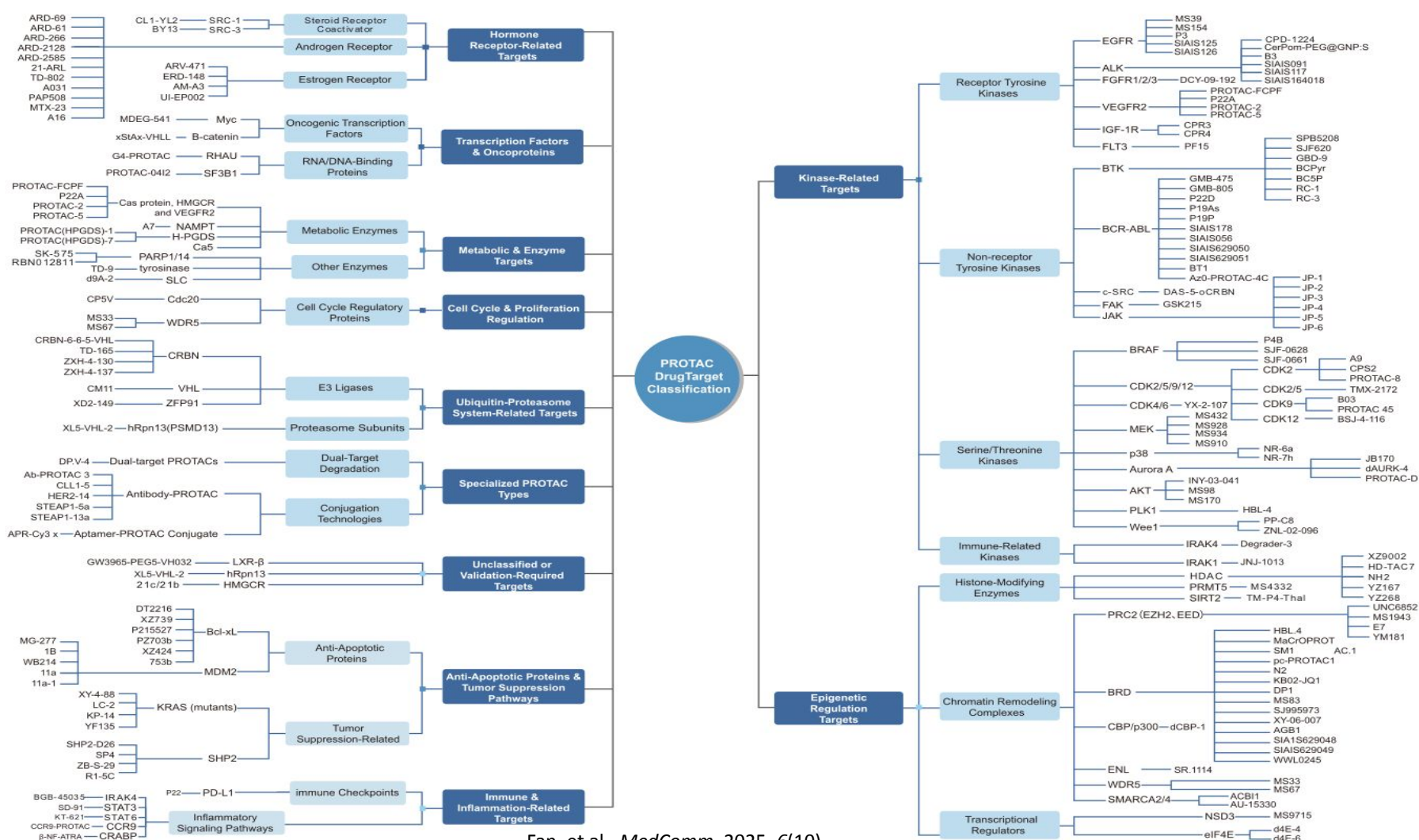
Vepdegestrant, a PROTAC **Estrogen** Receptor Degradar, in advanced **breast** Cancer.

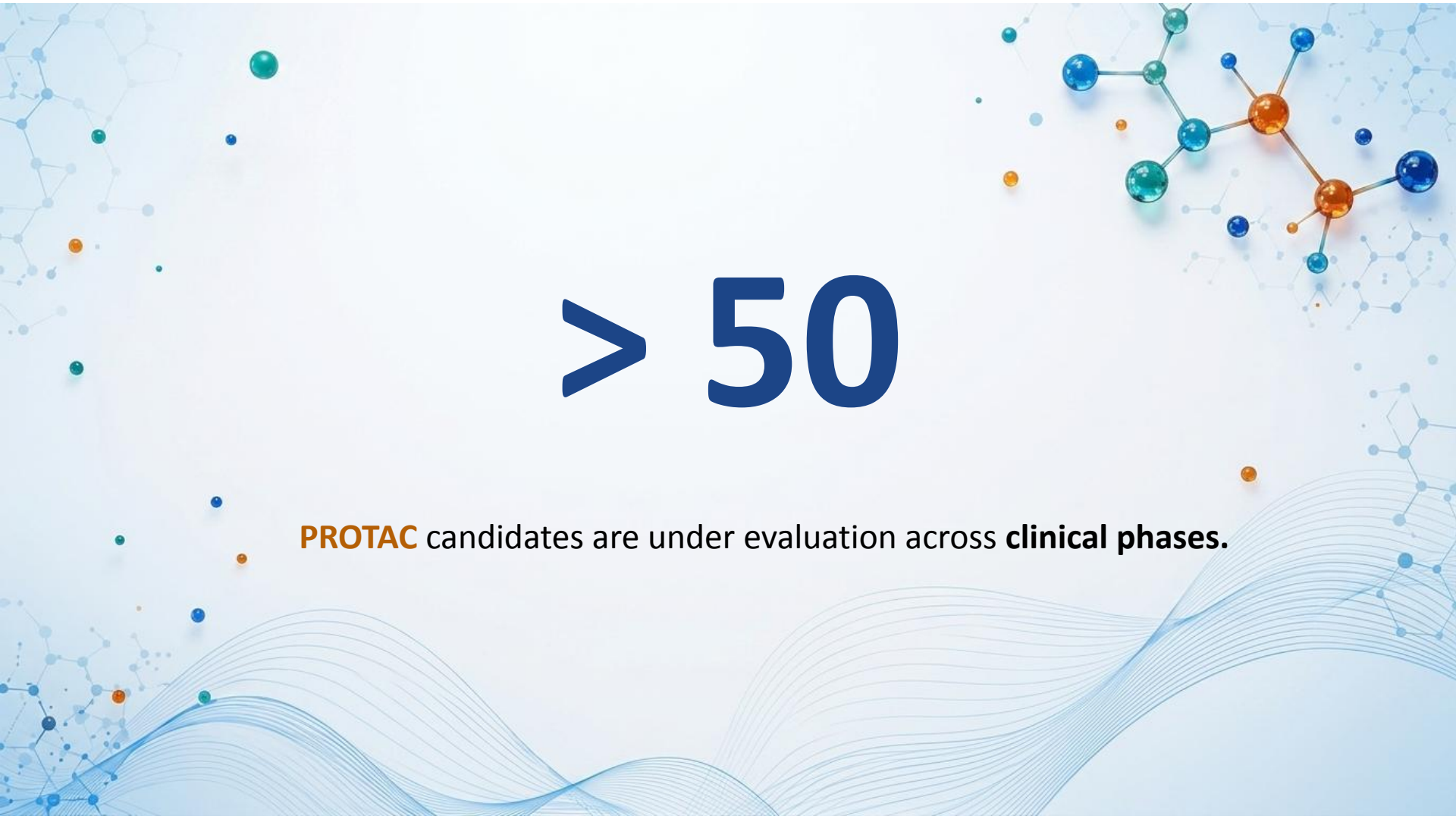


Examples of PROTACs

Company	Degrader	Target	Indications	E3 ligase	ROA	Highest phase
Arvinas	ARV-110	AR	Prostate cancer	CRBN	Oral	Phase II
Arvinas/Pfizer	ARV-471	ER	Breast cancer	CRBN	Oral	Phase II
Accutar Biotech	AC682	ER	Breast cancer	CRBN	Oral	Phase I
Arvinas	ARV-766	AR	Prostate cancer	Undisclosed	Oral	Phase I
Bristol Myers Squibb	CC-94676	AR	Prostate cancer	CRBN	Oral	Phase I
Dialectic Therapeutics	DT2216	BCL-x _L	Liquid and solid tumours	VHL	I.v.	Phase I
Foghorn Therapeutics	FHD-609	BRD9	Synovial sarcoma	Undisclosed	I.v.	Phase I
Kymera/Sanofi	KT-474	IRAK4	Autoimmune diseases (e.g., AD, HS, RA)	Undisclosed	Oral	Phase I
Kymera	KT-413	IRAK4	Diffuse large B cell lymphoma (MYD88-mutant)	CRBN	I.v.	Phase I
Kymera	KT-333	STAT3	Liquid and solid tumours	Undisclosed	Undisclosed	Phase I
Nurix Therapeutics	NX-2127	BTK	B cell malignancies	CRBN	Oral	Phase I
Nurix Therapeutics	NX-5948	BTK	B cell malignancies and autoimmune diseases	CRBN	Oral	Phase I
C4 Therapeutics	CFT8634	BRD9	Synovial sarcoma	CRBN	Oral	IND-e
C4 Therapeutics	CFT8919	EGFR-L858R	Non-small-cell lung cancer	CRBN	Oral	IND-e
Cullgen	CG001419	TRK	Cancer and other indications	CRBN	Oral	IND-e

Békés et al., Nat Rev Drug Discov, 2022, 21.



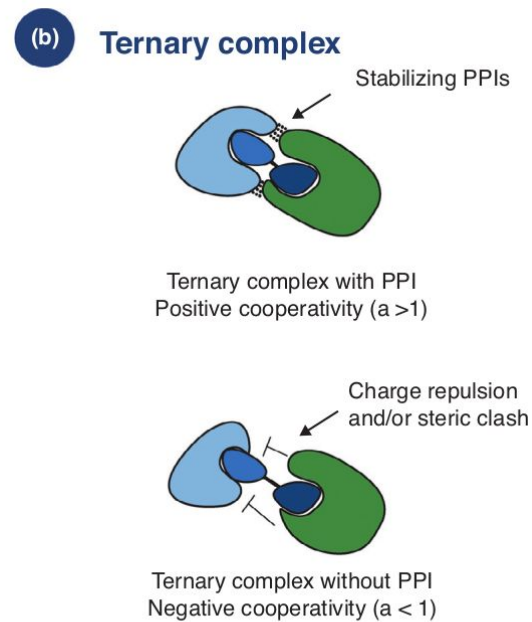
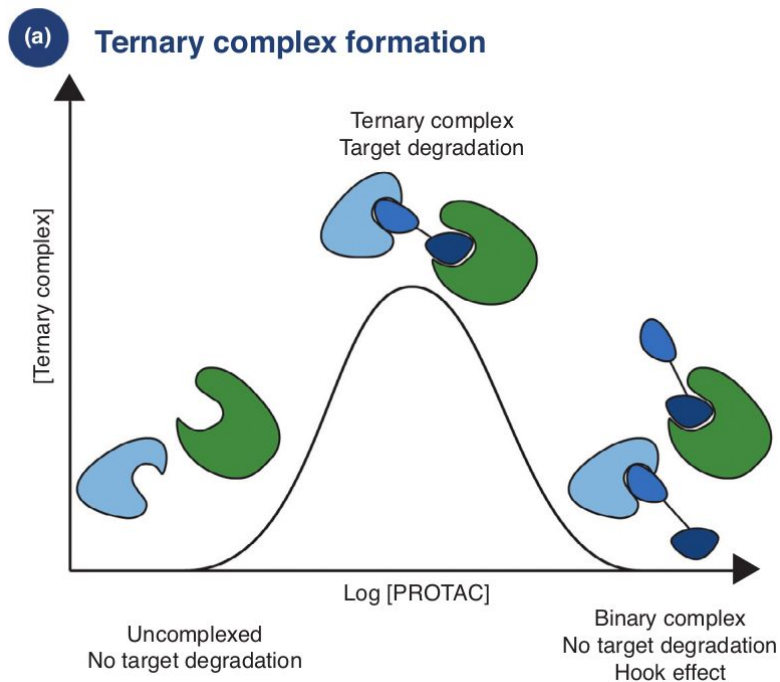


> 50

PROTAC candidates are under evaluation across **clinical phases**.

Factors affecting PROTACs

- ✓ **High-affinity binding** of both the POI ligand and the E3 ligand.
- ✓ The **stability and cooperativity** of the ternary complex.
- ✓ **Linker** properties (impact target selectivity).

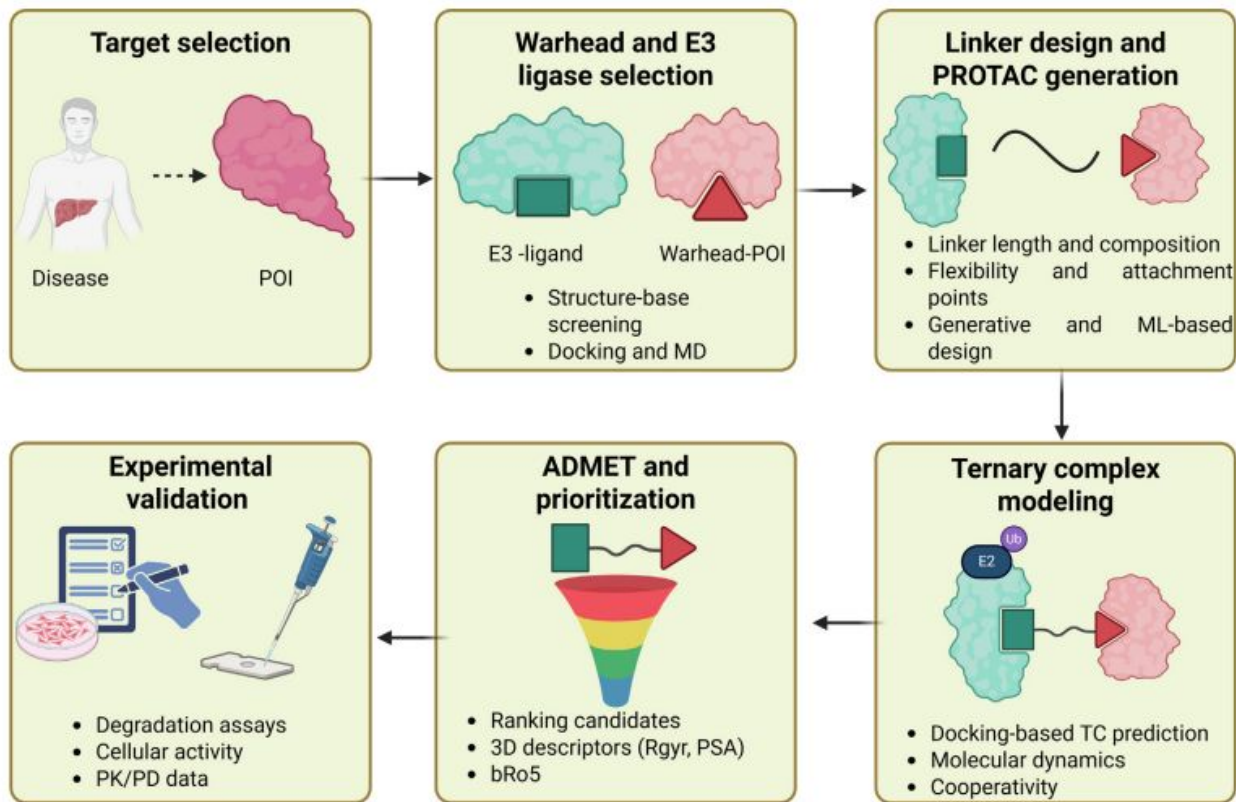


Toxicity of PROTAC

The potential toxicities of PROTACs are **less severe** than those of traditional on-target inhibitors because they are **event-driven** technology.

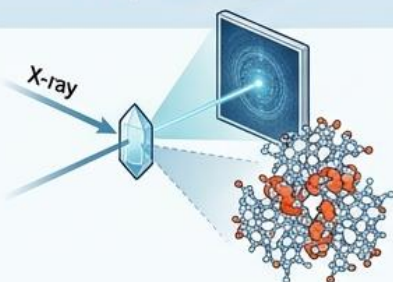
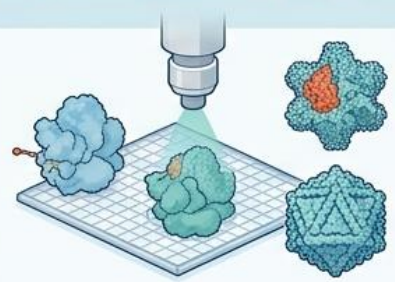
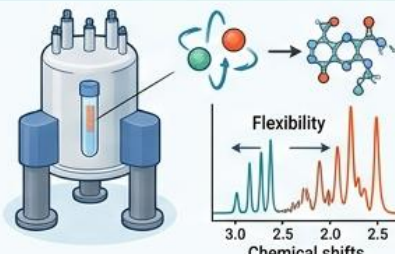
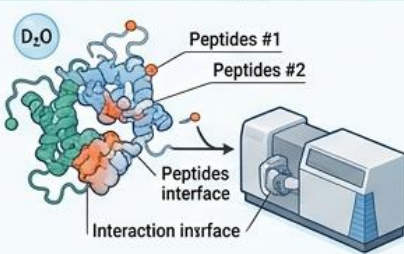
The background features a light blue gradient with abstract molecular structures. In the top right, there is a prominent cluster of spheres (blue, green, and orange) connected by lines, resembling a chemical molecule. The bottom of the image is decorated with several thin, overlapping, wavy blue lines that create a sense of movement and depth.

Development of PROTACs

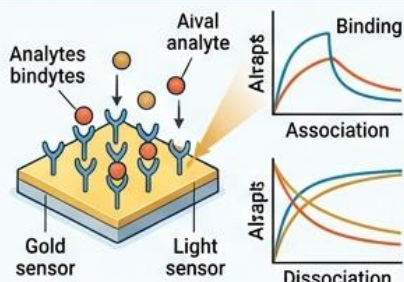
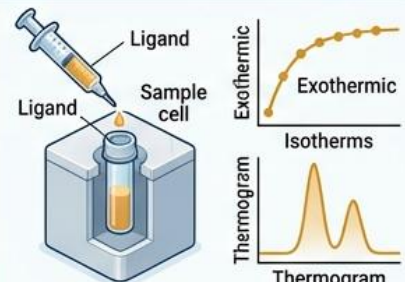
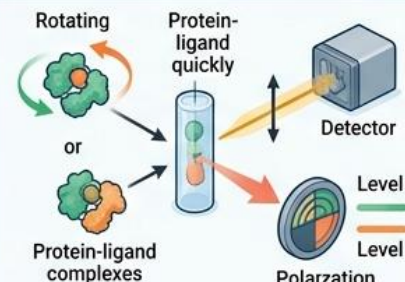
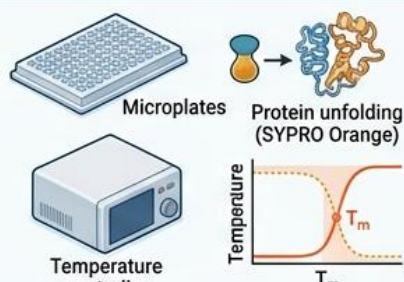


Drug Discovery Today

STRUCTURAL BIOLOGY TOOLS

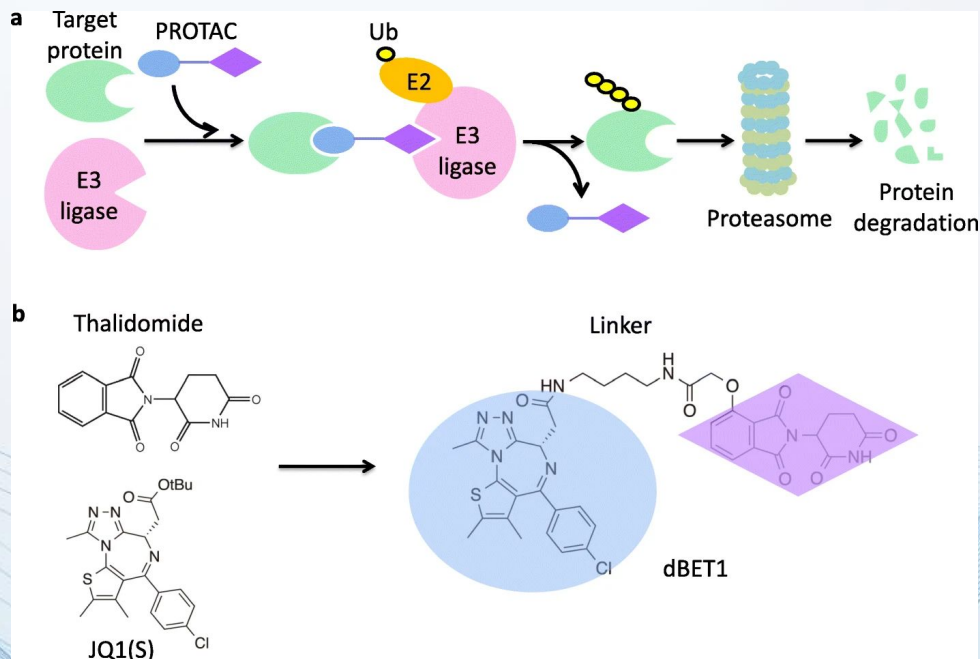
TECHNIQUE		WHAT IT REVEALS		STRENGTH			
X-ray Crystallography		Cryo-Electron Microscopy (Cryo-EM)		NMR Spectroscopy		Hydrogen-Deuterium Exchange Mass Spectrometry (HDX-MS)	
							
Atomic structure of ternary complexes.	High resolution	Structures of large or flexible complexes.	Near-native state	Protein flexibility and dynamics.	Dynamic information	Interaction interfaces and conformational changes.	Protein dynamics

BIOPHYSICAL METHODS

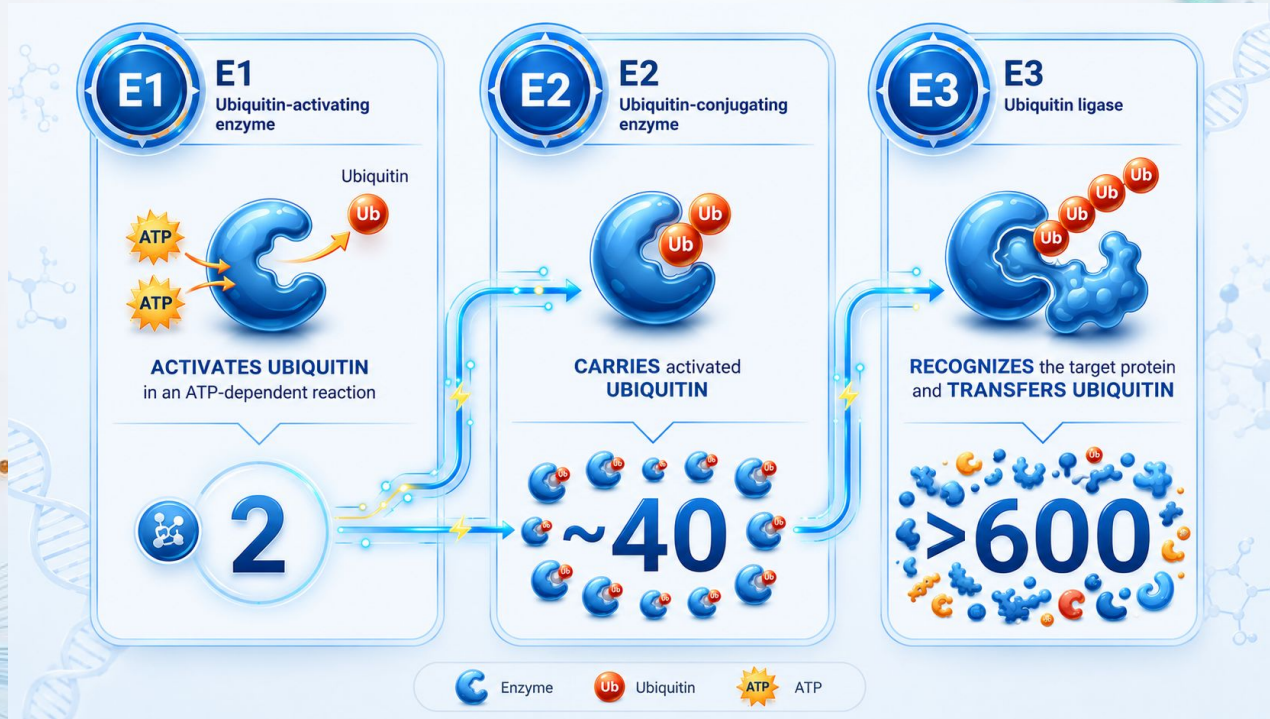
Surface Plasmon Resonance (SPR)  Binding kinetics	Isothermal Titration Calorimetry (ITC)  Binding affinity and thermodynamics	Fluorescence Polarization (FP)  Ligand binding	Differential Scanning Fluorimetry (DSF)  Protein stability
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Immunomodulatory imide drugs (IMiDs)

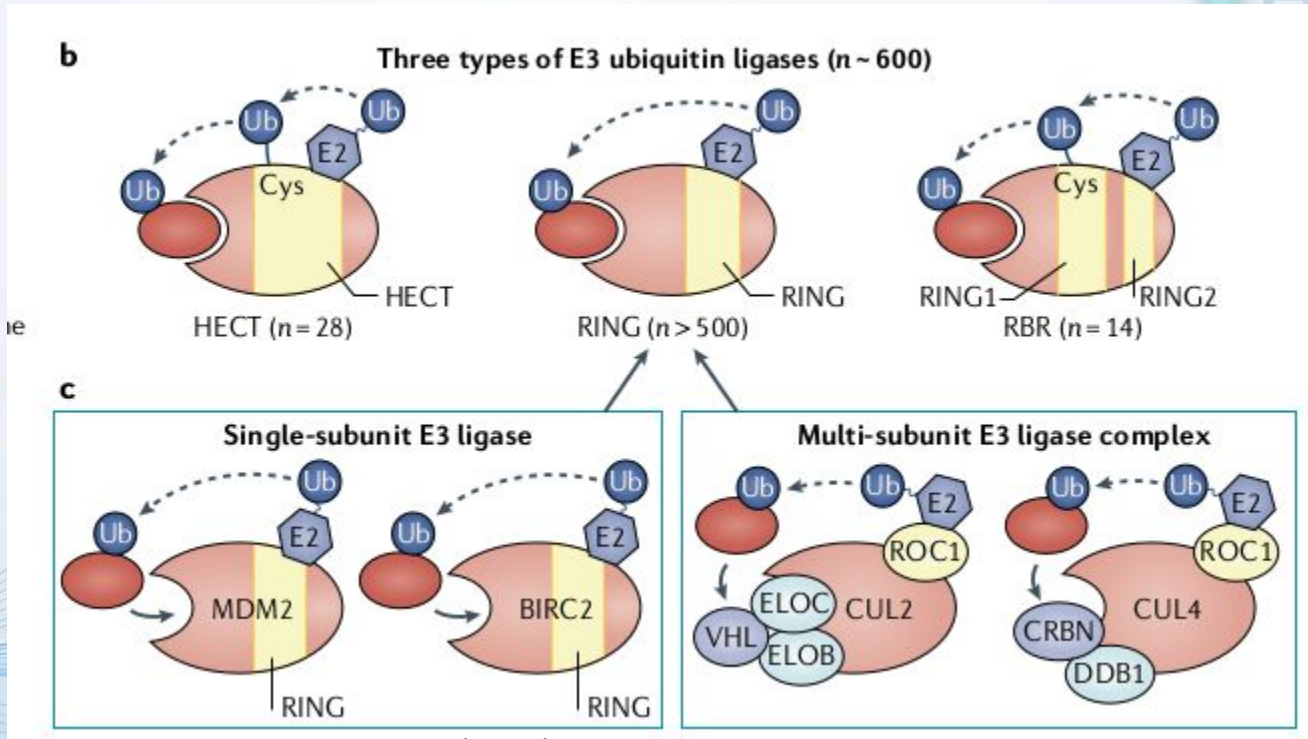
A class of small molecules have often been used as **E3 ligands** in PROTACs.



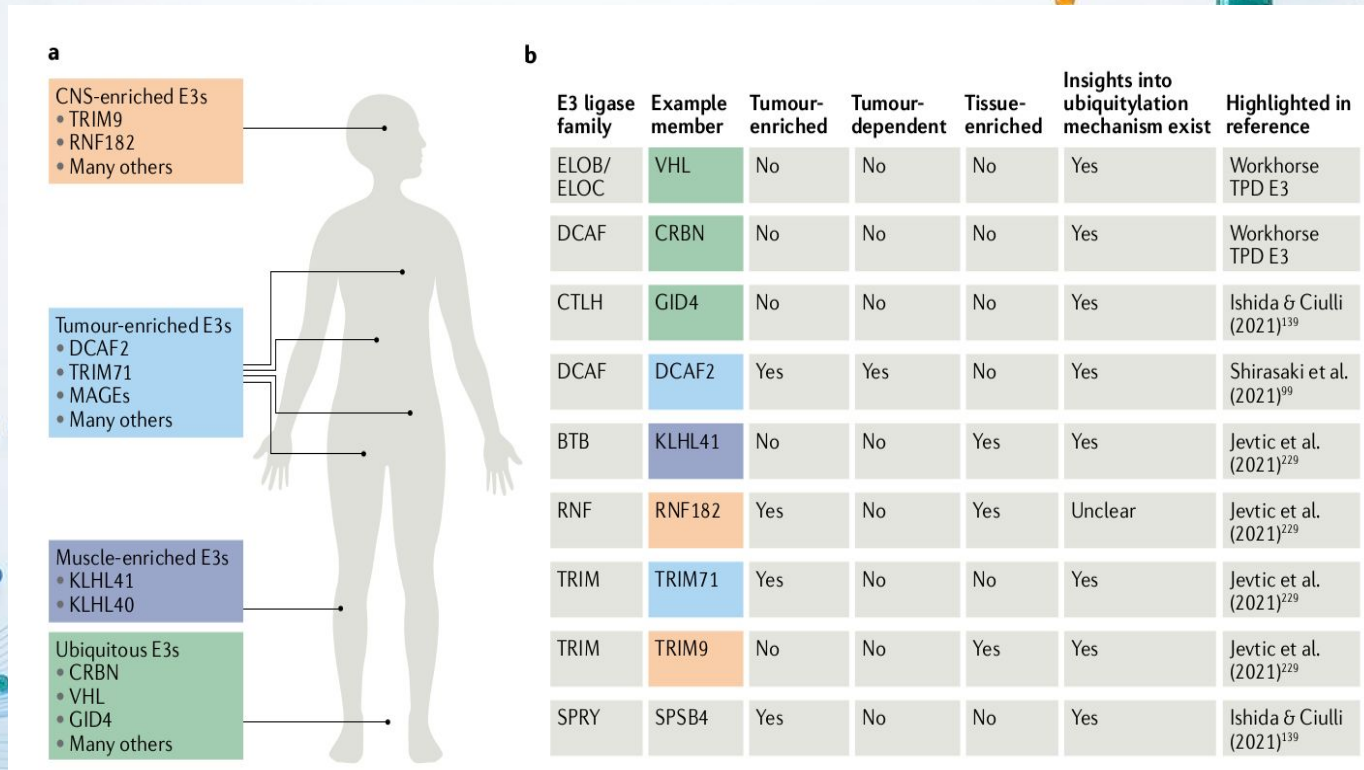
E1, E2, and E3 in human



Types of E3 ligases



Specialized E3 ligases



E3 ligands

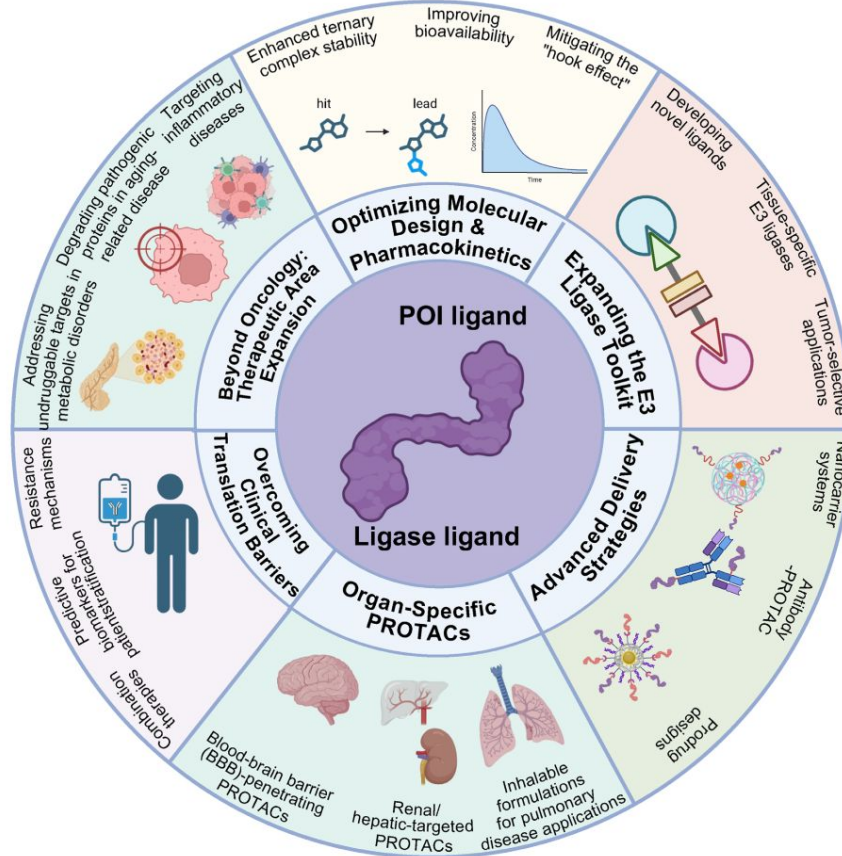
E3 ligand	Binding E3 subunit	E3 ligase complex ^a	Neosubstrates of the E3 ligand	Toxicity of E3 ligand	Mechanism of action	Functional essentiality of the E3 subunit		Refs
						In cells ^b	In mice	
Nutlin 3, idasanutlin	MDM2	MDM2	Not reported	High	PROTAC	5.2%	Embryonic lethal	171–173
VHL ligands	VHL	CRL2 ^{VHL}	Not reported	NA	PROTAC	0.5%	Embryonic lethal	12,33,89, 128,174–176
IMiDs	CRBN	CRL4 ^{CRBN}	ARID2, CK1 α , GSPT1, IKZF1, IKZF3, p63, ZBTB16, SALL4, ZFP91, ZNF276, ZNF653, ZNF692, ZNF827	High	Molecular glue and PROTAC	0%	Viable	7,9,75,77, 78,177–181
Bestatin, LCL161, MV-1	BIRC2	BIRC2	Not reported	Low	PROTAC	2.3%	Viable	182–189
Sulfonamides	DCAF15	CRL4 ^{DCAF1}	RBM39, RBM23	Medium	Molecular glue and PROTAC	NA	NA	79–82,129, 190–192
CCW16	RNF4	RNF4	Not reported	NA	PROTAC	2.3%	Embryonic lethal	132
KB02	DCAF16	CRL4 ^{DCAF1}	Not reported	NA	PROTAC	NA	NA	131
Nimbolide	RNF114	RNF14	Not reported	None	PROTAC	NA	Viable	193
CR8, HQ461, dCeMM2, dCeMM3, dCeMM4	DDB1	CRL4 ^{CDK12}	Cyclin K	NA	Molecular glue	67.2%	Embryonic lethal	83–85
Manumycin polyketides	UBR7	UBR7	p53	NA	Molecular glue	0%	NA	86

Dale et al., Nat Rev Cancer, 2021, 21.

Current challenges

- Molecular weight and polarity constraints limit oral bioavailability.
- Hook effect complicates dose optimization
- Resistance mechanism.
- Delivery and Distribution.

Future Directions and Improvement Strategies for PROTAC Technology



PROTAC database

PROTAC-DB

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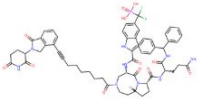
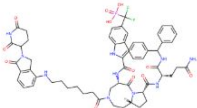
PROTACs ▾

Q

Showing 1 to 50 of 70 entriesPer page: 50 ▾

Filters

12Next >Last >>

ID/Target	Compound	Degradation		Protac to Target		Protac to E3		Ternary complex		Cellular activities (nM)			Paper
		DC50	Dmax	IC50	Kd	IC50	Kd	IC50	Kd	IC50	EC50	GI50	
478	 (Pred)	90								25			J Med Chem. 2019 Dec 26;62(24):11280-11300
STAT3		Degradation of total STAT3 and p-STAT3Y705 proteins in Molm-16 cells after 4 h treatment								Inhibit growth of Molm-16 cells			
489		90								59			J Med Chem. 2019 Dec 26;62(24):11280-11300
STAT3		Degradation of total STAT3 and p-STAT3Y705 proteins in Molm-16 cells after 4 h treatment								Inhibit growth of Molm-16 cells			

<https://cadd.zju.edu.cn/protacdb/>

Beyond PROTACs

Modality	Degradation Pathway	Key Advantage
 LYTACs	Lysosome	Degrades extracellular and membrane proteins
 AUTACs	Autophagy	Targets intracellular proteins and organelles
 ATTECs	Autophagosome	Tethers targets directly to LC3 for degradation
 AUTOTACs	Autophagy	Targets misfolded and aggregate proteins

References

Fan, G., Chen, S., Zhang, Q., Yu, N., Shen, Z., Liu, Z., Guo, W., Tang, Z., Yang, J., & Liu, M. (2025). Proteolysis-Targeting Chimera (PROTAC): Current Applications and Future Directions. *MedComm*, 6(10), e70401. <https://doi.org/10.1002/mco2.70401>

Pettersson, M., & Crews, C. M. (2019). PROteolysis TArgeting Chimeras (PROTACs) — Past, present and future. *Drug Discovery Today Technologies*, 31, 15–27. <https://doi.org/10.1016/j.ddtec.2019.01.002>

Dale, B., Cheng, M., Park, K., Kaniskan, H. Ü., Xiong, Y., & Jin, J. (2021). Advancing targeted protein degradation for cancer therapy. *Nature Reviews. Cancer*, 21(10), 638–654. <https://doi.org/10.1038/s41568-021-00365-x>

Békés, M., Langley, D. R., & Crews, C. M. (2022). PROTAC targeted protein degraders: the past is prologue. *Nature Reviews Drug Discovery*, 21(3), 181–200. <https://doi.org/10.1038/s41573-021-00371-6>

The background is a light blue gradient. It features several abstract elements: in the top right, a cluster of 3D molecular models with blue, green, and orange spheres connected by lines; in the bottom, a series of thin, wavy blue lines that create a sense of movement; and scattered throughout are various small, semi-transparent molecular structures and individual colored spheres (blue, green, orange).

Thank you